

Notional Institute of Technology, Kurukshetra

BTech(4th Sem), S-1

Subject & code: Computer Networks (CSPC-202 &ITPC-202)

D.O.E: 19/02/2026

Time: 50 mts

Marks: 15

Attempt all the questions. All question carry equal marks.

Q1.	Describes IEEE 802.3 and IEEE 802.11 standards in detail in view of Medium, Mobility, Topology, Access Method, Flexibility, Speed and Security under Physical and Data Link Layer.	(5)
Q2.	The following character encoding is used in a data link protocol: A: 01000111 B: 11100011 FLAG: 01111110 ESC: 11100000 Show the bit sequence transmitted (in binary) for the four-character frame A B ESC FLAG when each of the following framing methods is used: - Byte count. - Flag bytes with byte stuffing. - Starting and ending flag bytes with bit stuffing.	(1, 2, 2)
Q3.	Consider the diagram shown below where a number of LANs are connected by (transparent) bridges(B1, B2, B3, B4, B5). Here, H represents the Host(or Machine) connected with the LANs. In order to avoid packets looping through circuits in the graph, the bridges organize themselves in a spanning tree. First, the root bridge is identified as the bridge with the least serial number. Next, the root sends out (one or more) data units to enable the setting up of the spanning tree of shortest paths from the root bridge to each bridge. Each bridge identifies a port (the root port) through which it will forward frames to the root bridge. Port conflicts are always resolved in favour of the port with the lower index value. When there is a possibility of multiple bridges forwarding to the same LAN (but not through the root port), ties are broken as follows: bridges closest to the root get preference and between such bridges, the one with the lowest serial number is preferred. For the given connection of LANs by bridges; - Find out the bridge tables(Consider bridge name as MAC address) - Identify root port, designated port and blocked port - Find out the depth first traversal of the spanning tree of bridges	(2, 2, 1)